SIDEBAR

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**There are in fact research papers in Bayesian optimization that seek to search for the location of the optimizer x* using entropy. The resulting policies, however, often prove too computationally expensive to run and are not implemented in BoTorch.**

But that doesn't mean we will have to abandon the effort of using information theory in BayesOpt altogether. It just means that we need to modify our search procedure to make it more amenable to computational methodologies. One easy way to achieve this is to target a quantity other than the optimizer $x^*$ that is, on one hand, connected to the search for $x^*$, and on the other, easier to reason about.

A quantity of interest in optimization, other than the optimizer $x^*$, is the optimal value $f^* = f(x^*)$, achieved at the optimizer. As one can imagine, learning about the optimal value $f^*$ might tells us a lot about the optimizer $x^*$; that is, the two quantities are information-theoretically connected. However, the optimal value $f^*$ is much easier to work with than the optimizer $x^*$, as the former is just a real-valued number, while the latter is a vector of length equal to the number of dimensions of the objective function.

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